

Multi-Agent Architecture

<https://github.com/aiyshwary/multi-agent-revenue-churn>

OVERVIEW

A production-grade agentic pipeline that combines LLM-driven planning with deterministic Python analytics for revenue and churn prediction, featuring an Orchestrator-Executor-Reflection loop with memory management and Docker.

IMPACT

Designed and implemented a multi-agent revenue and churn analysis system where an LLM Planner coordinates an Orchestrator, Executor, and Reflection agent, keeping LLMs focused on reasoning while Python handles all numeric computation.

KEY DETAILS

1. Planner emits an executable plan (list or graph of nodes) from a high-level task description.
2. Orchestrator runs each step via the Executor, applies deterministic validators, and uses Reflection for iterative improvement.
3. Memory Manager persists short summaries and semantic snippets for context retrieval across long sessions.
4. Outputs churn_report.json, per-client quarterly revenue CSVs, and HTML/PNG visualizations per report.
5. Fully containerized with Docker Compose; CI/CD via GitHub Actions with pytest unit test suite.

6. GraphRunner resolves step dependencies as a DAG and executes ready nodes in parallel via ThreadingLocalExecutor, CircuitBreaker and SHA-256 idempotency keys to prevent duplicate execution.

7. SemanticMemory uses a dependency-free in-memory cosine-similarity index with deterministic MD5 hashes for reproducible tests; long-term memory uses zlib compression and persists to JSON.

8. LLMClient supports both a mock provider (deterministic stubs for offline testing) and OpenAI provider; it enforces budgets before every call. Visualizations generated with Matplotlib (PNG) and Plotly (interactive).